1 Human Body: The Circulatory System

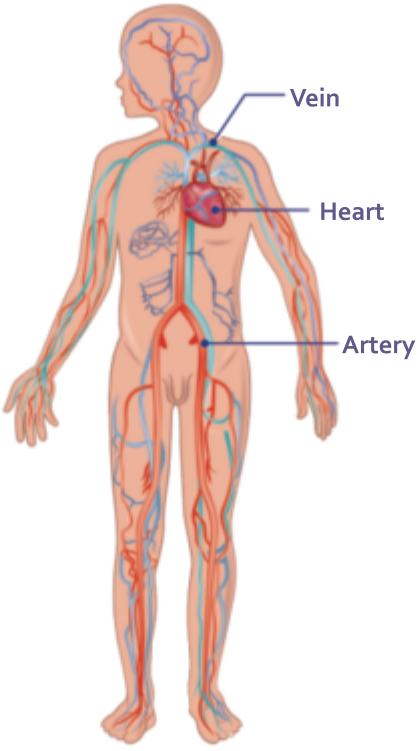
Learn about

- The circulatory system
- Heart
- Blood
- Blood vessels
- Process of circulation

We know that a group of organs[1] working together to perform a particular function is called an organ system. For example, the digestive system helps us to break down the food we eat into simpler forms that can be used by our body. The mouth, food pipe, stomach, small intestine, large intestine, and anus together form the digestive system. Similarly, the excretory system removes wastes, such as urine and sweat, from the body. Kidneys, skin, and lungs form the excretory system. We will learn about the organ system called the circulatory system in this chapter.

▶ The Circulatory System

*The **circulatory system** is an organ system that helps in transporting substances such as nutrients, water, and oxygen inside our body, and also removes waste gases, such as carbon dioxide, from the body through blood.*



Circulatory system

Parts of the Circulatory System

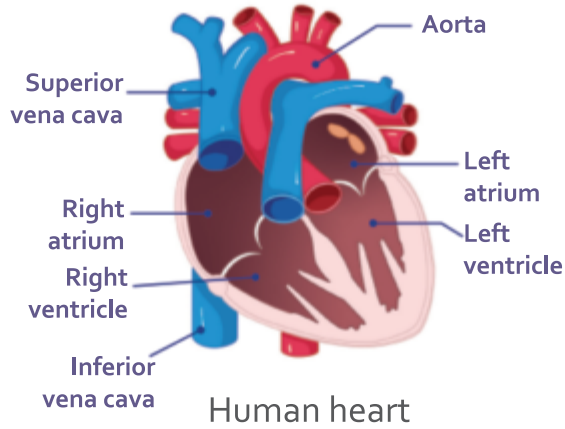
The circulatory system is made up of the following three main parts:

- Heart
- Blood
- Blood vessels

[1]**organ:** a part of the body that carries out a specific function

▶ Heart

*The **heart** is a muscular organ that pumps blood to all parts of our body.* It is located between the lungs, slightly to the left of the middle of the chest. It is protected by the ribcage.



Human heart

Structure and Function of the Heart

The human heart is a muscular organ consisting of four chambers—two upper chambers and two lower chambers. The upper chambers are called **atria** or **auricles**. They receive blood coming from the rest of the body. The lower chambers are called **ventricles**. They pump blood out of the heart. The adult human heart is about the size of our fist.

The main function of the heart is to pump blood containing oxygen and nutrients to all cells in the body. It also receives blood containing carbon dioxide from all the parts of the body and pumps it to the lungs for purification.

*The rhythmic movement that the heart makes while pumping blood is called the **heartbeat**.* Let us perform an activity.

SCIENTIFIC SKILLS · EXPERIENTIAL LEARNING

Activity

Aim: To listen to our heartbeat

Materials required: Plastic funnel attached to a rubber tube (about half metre long), and a rubber sheet (you can cut a large balloon and use it)

Procedure:

1. Stretch the rubber sheet tightly over the wide side of the funnel and secure it with a rubber band. Your heartbeat listening kit is ready.
2. Now, place the mouth of the funnel on your chest, and listen to the heartbeat from the other end of the tube. Do you hear any sound?

Observation: After some practice, you will be able to hear a thumping sound.

Conclusion: This thumping sound is our heartbeat. Normally, the heart beats 72 times per minute in an adult.

We can feel the rhythmic movement that the heart makes at different places on our body, such as the wrist or the neck. Let us learn how to do it.

Activity

Aim: To feel the pulse at our left wrist position

Procedure:

1. Turn your left palm upward.
2. Place two fingers of your right hand on your left wrist, as shown in the figure.
3. You will feel a light pumping movement. This is called a **pulse**. For an adult human being, the pulse rate can vary from 60 to 100 beats per minute.

For our body to function properly, it is important to keep our heart healthy. The best way to do this is to adopt a healthy and active lifestyle. We should do the following regularly:

- Walk daily for at least 3 to 4 km, ideally in the morning.
- Exercise for at least 30 minutes every day in the morning.
- Ask your parents to let you walk or cycle to school if it is a short distance.
- Avoid sitting for too long. Get up at regular intervals, walk around, and then return to sitting.
- Actively participate in outdoor sports such as football, cricket, and badminton.
- Always eat a balanced diet.
- Eat less junk and fried food as they are bad for the heart. Aim for an ideal body weight.
- Avoid polluted surroundings and stay away from cigarette smoke.

▶ Blood

Blood *is a substance that flows inside the human body through muscular tubes called blood vessels.* Blood is pumped by the heart to different parts of the body.

The Composition of Blood

Blood consists of (a) fluid part known as plasma and (b) three kinds of blood cells:

- Red blood cells (RBCs)
- White blood cells (WBCs)
- Blood platelets

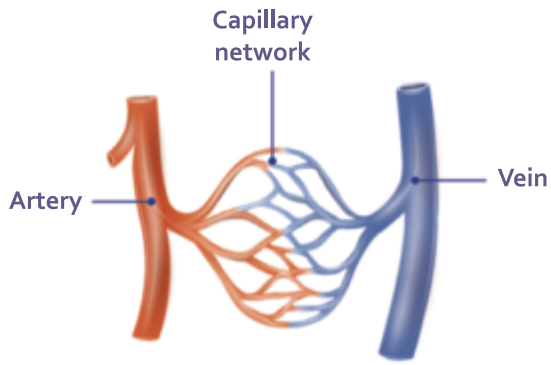
Blood is primarily responsible for the transportation of nutrients (from the digested food), water, oxygen, and carbon dioxide within our body.

Functions of Blood

- Blood helps in transporting oxygen and carbon dioxide throughout the body. The RBCs present in the blood take the oxygen from the lungs to all the cells in the body. They also collect carbon dioxide from these cells, and take it back to the lungs.
- Blood carries the nutrients generated during digestion from different parts of the digestive system to all the cells of the body.
- Blood helps in carrying the waste generated by the cells of the body to the kidneys for filtering.
- Blood helps in maintaining the temperature of the body, that is, it prevents the body from becoming too hot or too cold.
- The WBCs present in the blood protect the body from infections and disease-causing organisms.

▶ Blood Vessels

Blood vessels *are thin muscular tubes that carry blood throughout our body.* They form a wide network within the body so that blood can be transported to all the parts of the body. Blood vessels are of three types: arteries, veins, and capillaries.



Blood vessels

Structure and Functions of Blood Vessels

Let us study about the different types of blood vessels in our body.

Arteries

Arteries have the thickest walls of the three blood vessels. They carry blood containing oxygen from the heart to different parts of the body. Each artery gets divided into smaller **arterioles** to form a network throughout the body.

The largest artery in our body is called the **aorta**. It is the also the largest blood vessel in our body. It is the main artery coming out of the heart. Another important artery is the **carotid artery**, which carries blood from the heart to the brain.

Veins

Veins have thinner walls than arteries and carry blood from different parts of the body back to the heart. Veins can carry more blood than the arteries. They contain special structures called **valves** that prevent blood (that is going towards the heart) from flowing back to the body parts. Veins divide into smaller **venules** to form a network throughout the body.

Important veins of the human body include the **superior vena cava** (which brings blood from the upper body to the heart) and the **inferior vena cava** (which brings blood from the lower body to the heart).

Capillaries

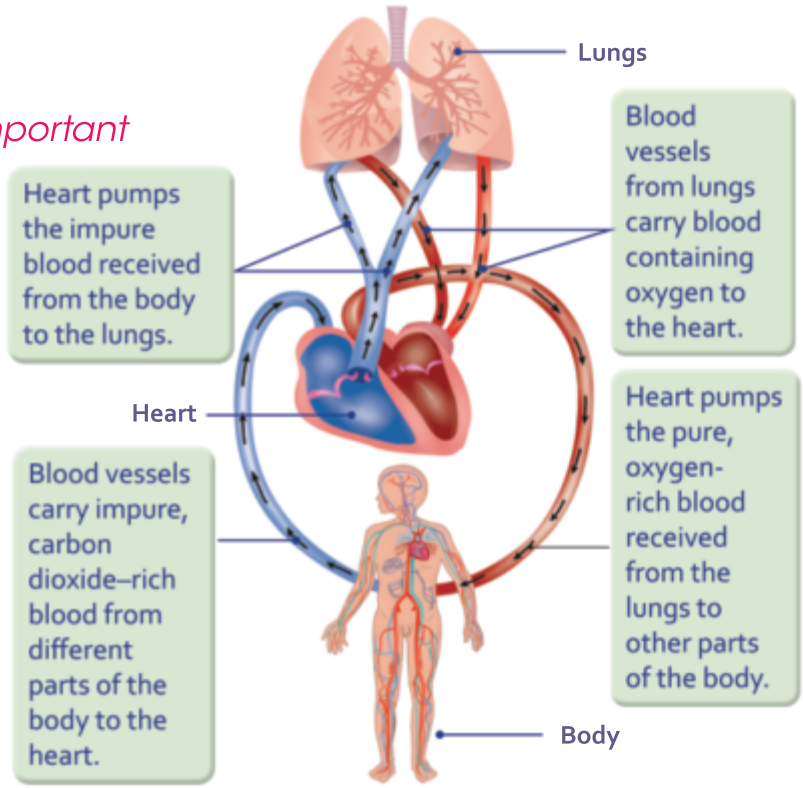
Capillaries are the thinnest of the three blood vessels. They connect the arteries to the veins.

▶ Process of Circulation

*The continuous movement of important nutrients, oxygen, and carbon dioxide between the heart and the rest of the body by blood through the blood vessels is called **circulation**.*

Fact File

When the body is at rest, it takes only six seconds for the blood to travel from the heart to the lungs and back.



Circulation of blood

Questions

Choose the correct option to fill in the blank.

1. The circulatory system is the name of an ..... (organ/organ system).
2. The upper chambers of the heart are called ..... (auricles/ventricles).

3. ..... (Superior/Inferior) vena cava brings blood from the lower body to the heart.
4. ..... (Atria/Ventricles) are the upper chambers of the heart.

Yoga Exercises for a Healthy Heart

Yoga is a good form of exercise to keep the heart healthy and to improve the circulation of blood. Regular practice of simple yoga exercises improves the overall fitness of the body. There are several benefits of practising yoga.

- It increases muscle strength and improves the function of lungs.
- It helps in decreasing stress and anxiety and gives a feeling of well-being.
- It boosts energy, improves memory, and increases concentration.
- It helps in reducing weight and also prevents high blood pressure and high blood sugar level.
- It helps relieve joint pain in elderly persons.

Some yoga exercises that may improve heart health are **tadasana** (mountain pose), **vrikshasana** (tree pose), **trikonasana** (triangle pose), and **dhanurasana** (bow pose). Yoga may also be combined with meditation, which helps clear the mind of negative thoughts and reduces stress. Deep breathing exercises improve lung function and blood circulation. Yoga exercises should always be done under the guidance of an expert or a trained yoga teacher.

Tadasana

Vrikshasana

Trikonasana

Dhanurasana

Wrap Up

- The circulatory system is made up of the heart, blood, and blood vessels.
- The heart is a muscular organ that pumps blood to all parts of our body.
- Blood flows inside the blood vessels and transports nutrients, water, oxygen, and carbon dioxide throughout the body.

- Blood vessels are thin tubes that carry blood throughout our body.
- Arteries carry blood from the heart to different parts of the body.
- Veins carry blood from the different parts of the body back to the heart.
- Capillaries connect arteries to veins. They are the thinnest of the blood vessels.
- The continuous movement of important nutrients, oxygen, and carbon dioxide between the heart and the rest of the body by blood through the blood vessels is called circulation.
- Doing simple yoga exercises under the guidance of an expert or a trained yoga teacher can keep our heart strong and healthy.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A Choose the correct option.

1. What do you call a group of organs working together to perform a particular function?
   a. A cell system b. A tissue system
   c. An organ system d. None of these
2. Which of these does the circulatory system transport?
   a. Oxygen b. Nutrients c. Water d. All of these
3. This is not a blood vessel.
   a. Atria b. Capillaries c. Arteries d. Veins
4. What should we practise to keep our heart healthy and strong?
   a. Balanced diet b. Exercise c. Yoga d. All of these
5. Which of these is the largest artery?
   a. Superior vena cava b. Aorta
   c. Arterioles d. Venules

B Assertion and Reasoning questions

1. **Assertion (A):** The circulatory system is responsible for transporting oxygen throughout the body.
   Reason (R): Blood carries red blood cells (RBCs) that are rich in oxygen.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

2. **Assertion (A):** Regularly checking your pulse helps monitor heart health.
   Reason (R): The pulse is a measure of your breathing rate.
   a. Both A and R are True
   b. Both A and R are False
   c. A is True and R is False
   d. A is False and R is True

C Choose the correct option to fill in the blank.

1. The heart is a ..... (muscular/non-muscular) organ that pumps blood to all parts of the body.
2. The lower chambers of the heart are called ..... (ventricles/auricles).
3. We should walk for at least half ..... (a minute/an hour) every day to keep our heart healthy.
4. The ..... (RBCs/WBCs) protect the body from infectious diseases.
5. The ..... (veins/capillaries) are the thinnest blood vessels.

SECTION II

D Short answer questions

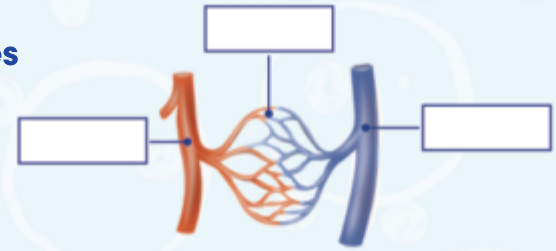
1. What is an organ system? Give an example.
2. Define heart. Also, give its location in the body.
3. Where can we feel the movement of our heartbeat in the body?
4. Maya loves french fries. She eats them almost every day after school. Her friend Varsha tries to convince her to eat more fruits and vegetables. How is Varsha correct?

E Long answer questions

1. Describe the structure and function of the heart.
2. A fitness instructor wants to teach the class ways to keep the hearts healthy. List five activities (besides exercise machines) that can be done almost anywhere to keep your heart healthy.
3. In what ways can practising yoga regularly make your lifestyle better?
4. Write any three functions of blood.

Picture Study

Look at the picture shown and label the boxes given. Then, answer the following questions.



1. Which among the three is the thickest blood vessel? What is its role?

   .....

2. Which is the thinnest of the three blood vessels? Mention its role.

   .....

My Learning Corner

A Think about

1. Rohan is a self-disciplined boy. Since his school is nearby, he prefers walking to it and reaches on time too. Is it the right practice?

B Try out

1. Use modelling clay to create a 3D model of the circulatory system. Use colours to represent parts (e.g., red for arteries, blue for veins), and label each part.
2. Paste pictures of the main parts of the circulatory system along with their names. Also, write the main function of each part in your Science scrapbook.

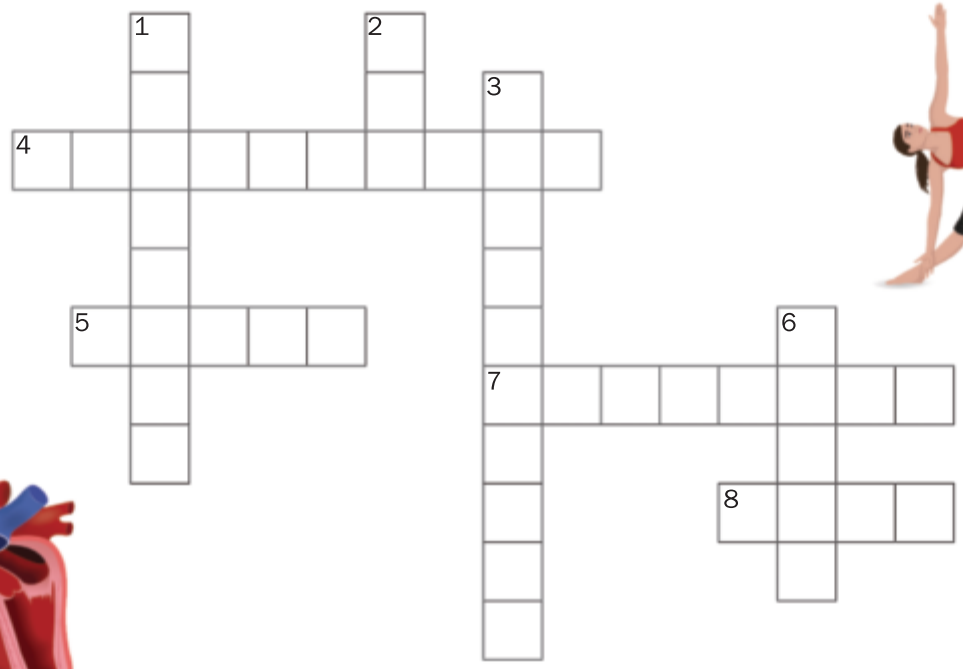
Self-Assessment

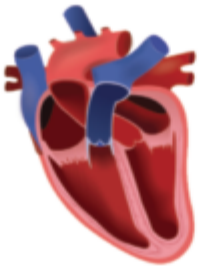
Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.
Score 5 = I can definitely do this.
Score 1 = I cannot do this yet.

I can...	My score
• identify organs of the circulatory system in a picture/model.	
• draw and label various organs of the circulatory system.	
• describe the functions of each organ and explain the process of circulation.	
• differentiate between arteries and veins and name the major arteries and veins.	
• explain the functions of blood.	
• discuss various ways to keep the heart healthy.	

Worksheet

Complete the following crossword with the help of the clues given below.





DOWN

1. A food item that is bad for the heart and should be avoided.
2. The cells present in the blood, which take oxygen from lungs to all the cells in the body.
3. By performing it, we clear our mind of all negative thoughts and de-stress ourselves.
6. A substance that is pumped by the heart, to different parts of the body.

ACROSS

4. The lower chambers in the heart.
5. The biggest artery in our body.
7. The upper chambers of heart.
8. A special form of exercise that keeps the heart healthy.